Assignment Instructions

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Q4-2

If the following liquid-phase reaction is elementry and reversible, what is the rate law in terms of conversion when equimolar amounts of A and B are fed to the reactor

$A \Leftrightarrow B$

Answer:

$$-r_A = k\left[C_A - \frac{C_B}{K_C}\right]$$

$$C_A = C_{A0}(1-x)$$

$$C_B = C_{A0}(\Theta_B + x)$$

$$\Theta_B = 1$$

$$-r_A = -k\left[C_{A0}(1-x) - \frac{C_{A0}(1+x)}{K_C}\right]$$

$$-r_A = kC_{A0}\left[(1-x - \frac{(1+x)}{K_C}\right]$$

This question is appropriate for an exam because it integrates reaction stoichiometry, equilibrium concepts and conversion-based rate laws which is very important in this class!

P4-1

A.1 As θ_B changes the reaction rate changes as well this is because $-r'_{SO_2}$ is dependent on the value of θ_B. As θ_B decreases $-r'_{SO_2}$ increases and as θ_B increases $-r'_{SO_2}$ decreases

A2 θ_B, $P_{SO_2,0}$ and k has the greatest effect on the reaction while K_{SO_3}, K_{O_2} and K_P has the least effect on the reaction rate.

A3

1. The reaction rate decreases rapidly with conversion, approching 0 near $X=0.7$ due to equilibrium limitations and SO_2 depletion, as clearly show by the shape of in the plot.

2. The parameters with the greatest influence are the rate constant k, the feed ratio and the equilibrium constant K_P, these directy affect the numerator driving force Adsorption parameters K_{O_2} and K_{SO_3} have comparatively minor effects

3. The curve demonstrates Langmuir-Hinshelwood behavior: at low conversion the forward kinetic term controls, while at higher conversion the reverse reaction and adsorption denominator dominate, causing the steep decline in rate

4. for reactor design, the plot shows that operating close to equilibrium would require extremly large catalyst weights; therefore practical operation must balance conversion with reasonable reactor size.

4 $F_{A0} = 3 \text{ mol}/h$

$X = 0.40$

$$\left.\frac{F_{A0}}{-r_A}\right|_{0.40} \approx 300 \text{ g-cat}$$

$$W = \left(\frac{F_{A0}}{-r_A}\right) X$$

$W = 300(0.4) = 120 \text{ g-cat}$

5a) $X = 0.30$

$$@ \quad \frac{F_A}{-r_A} \approx 50{,}000 \text{ g-cat}$$

$W_{30} = 50000\,(0.30) = 1.5 \times 10^4 \text{ g-cat}$

5b) 99% equilibrum

$X_{eq} = 0.75$

$X = 0.99\, X_{eq} = 0.74$

$$\frac{F_A}{-r_A} \approx 1.0 \times 10^6 \text{ g-cat}$$

$W = (1.0 \times 10^6)(0.74) = 7.4 \times 10^5 \text{ g-cat}$

1 PFR Volume for $F_{A0} = 5 dm^3/min$

$X = 0.40$

$$V = F_{A0} \int_0^{0.40} \frac{dx}{-r_A}$$

using trapazoidal rule

$$\int_0^{0.4} \frac{dx}{-r_A} \approx 0.1 \left[\tfrac{1}{2}(35+55) + \tfrac{1}{2}(55+80) + \tfrac{1}{2}(80+110) + \tfrac{1}{2}(110+150) \right]$$

$$= 0.1(337.5) = 33.8 \ dm^3 \cdot min/mol$$

$$V = F_{A0} \text{ (area)}$$

$$V = 5(33.8) = 169 \ dm^3$$

2. Batch time for 40% Conversion

$$t = C_{A0} \int_0^{0.40} \frac{dx}{-r_A}$$

$$C_{A0} = 0.05 \ mol/dm^3$$

$$t = 0.05(33.8) = 1.69 \ min$$

3. Large K_c, Small C_{A0} → X_{ef} farthest from X_{eb}

4. Small K_c, Large C_{A0} → X_{ef} Closet to X_{eb}

5. When K_c is set to 0.01 and C_{T0} is set to 0.005

X_{eb}/X_{ef} has a negative slope as y_{A0} apporaches 0

When K_c or C_{T0} is maximized The slope decreaces and is near

0. however when both values are maximized the slope goes back

to the orginal values.

6 Equilibrium Conversion controlled by K_c / C_{A0}; variable-volume effects are minor

at dilute feeds but grow with high reactant fraction

P4-3

$2A \rightleftarrows B$

feed A is fed at a concentration of 4 mol/dm³

Equilibrium conversion is found to be 60%

a) gas phase

$$C_x = (C_{x0} + \nu_x C_{A0} x)\left(\frac{1}{1+\varepsilon x} P \frac{T_0}{T}\right)$$

@ constant P and T

$$\varepsilon = -\frac{1}{2} \qquad \frac{\text{Change in total number of moles for complete conversion } \Delta\nu}{\text{mole of A reacted}}$$

$$C_A = \frac{F_A}{V} = \frac{F_{A0}(1-x)}{V} = C_{A0}\frac{(1-x)}{1+\varepsilon x} P\left(\frac{T_0}{T}\right)$$

$$C_B = \frac{C_{A0}\left(\theta_B + \frac{b}{a}x\right)}{1+\varepsilon x} P\left(\frac{T_0}{T}\right) \qquad \theta_B = 0 \text{ Pure feed A}$$

$$C_A = 4\frac{1-0.6}{1-\frac{1}{2}(0.6)} = 2.286$$

$$C_B = 4\frac{(1/2)(0.6)}{1-\frac{1}{2}(0.6)} = 1.714$$

$$K_C = \frac{C_B}{C_A^2} = \frac{1.714}{(2.286)^2} = 0.328 \text{ dm}^3/\text{mol}$$

b) Liquid - Phase : Constant volume, no expansion

$C_A = C_{A0}(1-x)$, $C_B = C_{A0} x$

$C_A = 4(1-0.6) = 1.6$

$C_B = 4(0.6) = 2.4$

$$K_c = \frac{2.4}{(1.6)^2} = 0.469 \, dm^3/mol$$

c Gas phase, isothermal, no ΔP

$k = 2 dm^6/mol \cdot s$

$K_c = 0.5 dm^3/mol$

$$-r_A = k\left(C_A^2 - \frac{C_B}{K_c}\right)$$

using part A for C_A^2 and C_B

$$-r_A = 2\left[\left(\frac{4(1-x)}{1-\frac{1}{2}x}\right)^2 - \frac{1}{0.5}\left(\frac{4x}{1-\frac{1}{2}x}\right)\right]$$

d.) Constant-volume batch reactor

$C_A = 4(1-x)$ $C_B = 4x$

using part b for C_A and C_B

$$-r_A = 2\left[(4(1-x))^2 - \frac{4x}{0.5}\right]$$

P 4-4

$2A + B \rightarrow C$

Isothermally PFR with no pressure drop

Feed equal molar A and B

$C_{A0} = 0.1 \text{ mol/dm}^3$

$$X = \frac{\text{moles of A reacted}}{\text{moles of A fed}}$$

A: $\frac{-2}{2} = -1$ $\qquad C_A = C_{A0}(1-x)$

B: $\frac{-1}{2} = -\frac{1}{2}$ $\qquad C_B = C_{B0} - \frac{1}{2} C_{A0} x$

C: $\frac{1}{2} = \frac{1}{2}$ $\qquad C_C = \frac{1}{2} C_{A0} X$

1. $C_{B0} = 0.1 \text{ mol/dm}^3$ Equal molar feed

2. C_A and C_C @ $X = 0.25$

   $C_A = 0.1(1-0.25) = 0.075 \text{ mol/dm}^3$

   $C_C = \frac{1}{2}(0.1)(0.25) = 0.0125 \text{ mol/dm}^3$

3. C_B @ $X = 0.25$

   $C_B = 0.1 - \frac{1}{2}(0.1)(0.25) = 0.0875 \text{ mol/dm}^3$

4. C_B @ $X = 1$

   $C_B = 0.1 - \frac{1}{2}(0.1)(1) = 0.05 \text{ mol/dm}^3$

5. Rate Relationsips $\qquad 2A + B \rightarrow C$

   $r_C = -\frac{1}{2} r_A$

   $r_C = 2 \text{ mol/min} \cdot \text{dm}^3$

   $r_A = -2(2 \text{ mol/min} \cdot \text{dm}^3) = -4 \text{ mol/min dm}^3$

6. Rate Law $-r_A(x)$

$$-r_A = kC_A^2 C_B$$

$$C_A = C_{A0}(1-x)$$

$$C_B = C_{A0}(1-\tfrac{1}{2}x)$$

$$-r_A = k\left[(C_{A0}(1-x))^2 \cdot (C_{A0}(1-\tfrac{1}{2}x))\right]$$

$$-r_A = kC_{A0}^3(1-x)^2(1-\tfrac{1}{2}x)$$

$$k = 2$$

$$C_{A0} = 2$$

$$-r_A = 2\cdot(2)^3(1-x)^2(1-\tfrac{1}{2}x)$$

$$-r_A = 16(1-x^2)(1-\tfrac{1}{2}x)$$

7 Rate @ $x = 0.5$

$$-r_A = 16(1-0.5)^2(1-\tfrac{1}{2}(0.5))$$

$$-r_A = 3 \text{ mol/dm}^3\cdot\text{s}$$

P4-5

1. $C_{A0} = 16.13$ mol/dm^3

$C_{B0} = 55.5$ mol/dm^3

Species	Intial	Change	Final
Ethylene oxide	C_{A0}	$-C_{A0}x$	$C_A = C_{A0}(1-x)$
water	C_{B0}	$-C_{A0}x$	$C_B = C_{B0} - C_{A0}x$
ethylene glycol	0	$+C_{A0}x$	$C_C = C_{A0}x$

$-r_A = k\,[C_{A0}(1-x)][C_{B0} - C_{A0}x]$

$-r_A = 0.1(16.13(1-x))(55.5 - 16.13x)$

CSTR @ $x = 0.9$

@ 300K $\tau = \dfrac{C_{A0}x}{-r_A} = \dfrac{(16.13)(0.9)}{6.61} =$ 2.19 s

@ 350 K, $k = 2.3$

$\tau = 0.095\,s$

Volumes

$\nu = 200\ L/s$

$V = \tau\nu$

$V_{300} = 439\ dm^3$

$V_{350} = 22\ dm^3$

2. $C_2H_6 \rightarrow C_2H_4 + H_2$

Pure ethane, 6 atm, 1100 K, $\Delta P = 0$

$A \rightarrow B + C$

$\Delta\nu = 1$

$\varepsilon = \dfrac{\Delta\nu}{A} = 1$

$$C_A = \frac{C_{A0}(1-x)}{1+x}$$

$$C_B = C_C = \frac{C_{A0}x}{1+x}$$

$$-r_A = kC_A$$

$$-r_A = \frac{kC_{A0}(1-x)}{1+x}$$

@ constant volume batch, no expansion

$$C_A = C_{A0}(1-x)$$

$$-r_A = kC_{A0}(1-x)$$

3 $C_2H_4 + \frac{1}{2}O_2 \rightarrow C_2H_4O$

Stoichiometric feed

6 atm, 260°C, $\Delta P = 0$

$$A + \frac{1}{2}B \rightarrow C$$

$\Delta\nu$: $1 - 1 - \frac{1}{2} = -\frac{1}{2}$

$$y_{A0} = \frac{1}{1.5} = \frac{2}{3}$$

$$\varepsilon = y_{A0}\frac{\Delta\nu}{a} = \frac{2}{3}\left(\frac{-1/2}{1}\right) = \frac{-1}{3}$$

Partial Pressures

$$P_A = P_0 \frac{1-x}{1-\frac{1}{3}x}$$

$$P_{O_2} = P_0 \frac{\frac{1}{2}-\frac{1}{2}x}{1-\frac{1}{3}x}$$

Elementry rate:

$$-r_A' = K P_A P_{O_2}$$

$$-r_A' = K P_0^2 \frac{(1-x)(\frac{1}{2}-\frac{1}{2}x)}{(1-\frac{1}{3}x)^2}$$

@ fluidized batch Same expression because no ΔP and same ε

@ PBR identical form Since $\Delta P = 0$

4. $C_6H_6 + 2H_2 \rightarrow C_6H_{10}$

Stoichiometric feed

6 atm, 170°C, $\nu_0 = 50 \, dm^3/min$

Rate Constant $= \frac{53 \text{ mol}}{\text{Kg min atm}^2}$

$E = 80 \text{ KJ/mol}$

$A + 2B \rightarrow C$

$\Delta \nu = 1 - 1 - 2 = -2$

$$y_{A0} = \frac{1}{3}$$

$$\varepsilon = \frac{\Delta \nu}{a} y_{A0} = \frac{-2}{1}\left(\frac{1}{3}\right) = -\frac{2}{3}$$

$$P_A = P_0 \frac{1-x}{1-\frac{2}{3}x}$$

$$P_B = P_0 \frac{2-2x}{1-\frac{2}{3}x}$$

Rate law elementry 3rd order

$$-r_A' = k_B P_A P_B^2$$

$$-r_A' = k_B P_0^3 \left[\frac{(1-x)(2-2x)^2}{(1-2/3x)^3} \right]$$

r' @ $x = 0.8$

$1-x = 0.2$

$2-2x = 0.4$

$1 - \frac{2}{3}x = 1 - 0.53 = 0.467$

$$\frac{(1-x)(2-2x)^2}{(1-2/3x)^3} = 0.3147$$

$P_0^3 = 6^3 = 216$

$-r_A' = k_B \, 67.97$

$$k(T) = 53 \exp\left[\frac{E}{R} \left(\frac{1}{300} - \frac{1}{T} \right) \right]$$

$$\frac{E}{R} = 9622$$

@ 170°C (443K)

$k_{443} = 1.10 \times 10^6$

$-r_A' = 7.48 \times 10^7$

@ 270°C (543K)

$-r_A' = 7.06 \times 10^8$

Fludied CSTR

$$W = \frac{F_{A0} X}{-r'_A}$$

$$F_{A0} = \frac{y_{A0} P_0 v_0}{RT}$$ @ 170°C (443K)

$$F_{A0} = \frac{\frac{1}{3}(6)(50)}{(0.08206)(443)} = 2.75 \text{ mol/min}$$

@ 170°C $\quad W = \dfrac{2.75(0.8)}{7.48 \times 10^{7}} = 2.9 \times 10^{-8}$ kg

@ 270°C $\quad W = \dfrac{2.75(0.8)}{7.06 \times 10^{8}} = 3.1 \times 10^{-9}$ kg